

Sec 2.2 – The Derivative at a Point

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

In Sec 2.1 we considered problems with a change in height divided by a change in time. It reminded us of “rise over run” or “delta y over delta x,” $\frac{\Delta y}{\Delta x}$, when we calculate the slope of a line segment.

We can use this idea for more general functions, not just those measuring height. We define this ratio, which is the **average** rate of change of f over the interval from a to $a+h$ as the **difference quotient**.

$$\text{Average rate of change of } f \text{ over the interval from } a \text{ to } a+h = \frac{f(a+h)-f(a)}{h}.$$

Building off the work we did in Chap 01 with Limits, we will consider the difference quotient as h gets smaller and smaller, or its limit as $h \rightarrow 0$:

This **instantaneous** rate of change of f at a is also called the **derivative of f at a** ,

The **derivative of f at a** , written $f'(a)$, is defined as

$$\text{Rate of change of } f \text{ at } a = f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

If the limit exists, then f is said to be **differentiable at a** .

The derivative gives

- The rate of change of f at a . (In particular, if $f(x)$ gives **position at time x** , $f'(a)$ is the **instantaneous velocity**.)
- The slope of the tangent line to the graph of f at $x = a$.

Scientists have determined that the volume of wood that grows on a square kilometer, in meters³, is given by:

$$V(r) = 0.2r^2 - 20r + 600, \text{ where } r \text{ is cm rainfall per year and } 60 \leq r \leq 120.$$

Partner up

- (1) Calculate the **average** rate of change of V with respect to r over the intervals $90 \leq r \leq 100$ and $100 \leq r \leq 110$.

$$\text{Average rate of change of volume for } 90 \leq r \leq 100 = \frac{V(100) - V(90)}{10} = \frac{600 - 420}{10} = 18 \text{ meter}^3 / \text{cm}.$$

$$\text{Average rate of change of volume for } 100 \leq r \leq 110 = \frac{V(110) - V(100)}{10} = \frac{820 - 600}{10} = 22 \text{ meter}^3 / \text{cm}.$$

So the average rate of change of the wood grown increases as the rainfall increases.

- (2) By choosing small values of h , estimate the **instantaneous** rate of change of V with respect to r at $r=100$ cm.

First, using $h=0.1$ and $h=-0.1$, the difference quotient is:

$$\frac{V(100.1) - V(100)}{0.1} = 20.02 \text{ m}^3 / \text{cm} \quad \text{and} \quad \frac{V(99.9) - V(100)}{-0.1} = 19.98 \text{ m}^3 / \text{cm}.$$

Next, using $h=0.01$ and $h=-0.01$, the difference quotient is:

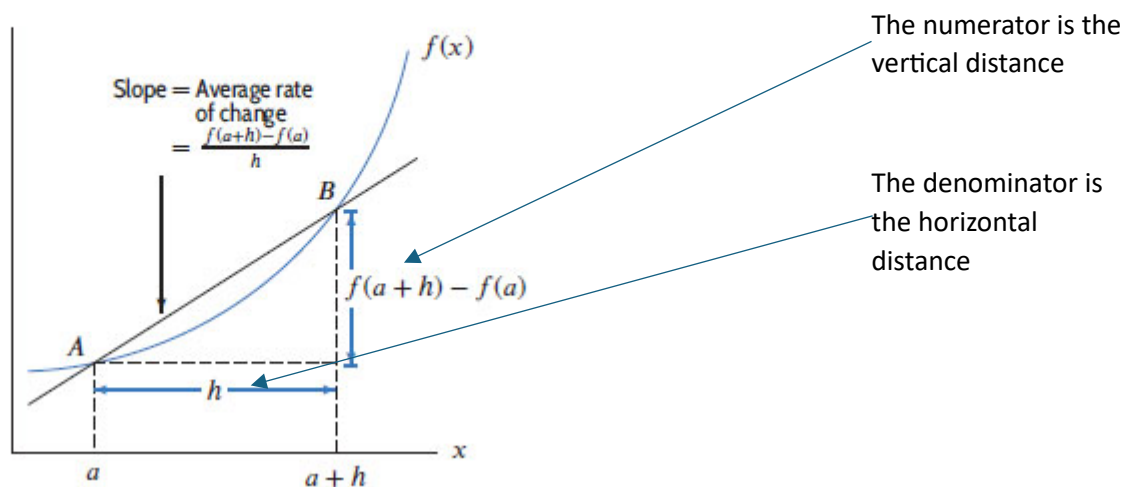
$$\frac{V(100.01)-V(100)}{0.01} = 20.002 \text{ m}^3/\text{cm} \quad \text{and} \quad \frac{V(99.9)-V(100)}{-0.1} = 19.998 \text{ m}^3/\text{cm}$$

These results suggest that when the annual rainfall is $r=100$ cm, the instantaneous rate of change of wood grown is about 20 meter^3 per cm of rainfall.

This second result is a numerical estimate. To get the exact rate of change of the difference quotient, ie. the derivative, we would need to calculate the limit exactly as $h \rightarrow 0$. Which we will do in a bit.

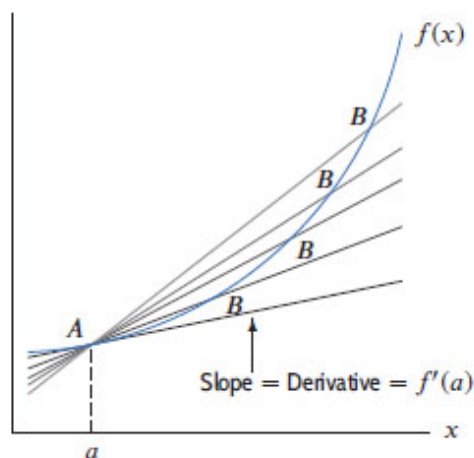
But first, let's consider a visualization of the derivative.

$$\text{Average rate of change of } f = \frac{f(a+h) - f(a)}{h} = \text{Slope of line } AB.$$



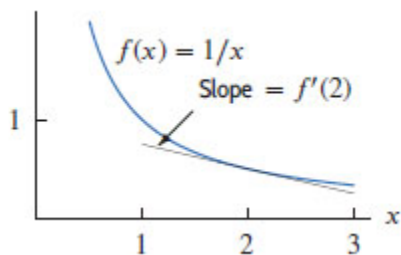
As h becomes smaller, the line AB approaches the tangent line to the curve at A .

$$\begin{array}{l} \text{Instantaneous rate of change} \\ \text{of } f \text{ at } a \end{array} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \text{Slope of tangent at } A.$$



Again we see as $h \rightarrow 0$, here graphically, that the derivative $f'(a)$ is the slope of the curve at A , or also the slope of the line tangent to the curve at A .

Here is a graph of $f(x) = \frac{1}{x}$



Partner up

What do you expect the sign of $f'(2)$ to be? That is, is the derivative of f at $x=2$ positive or negative?

Using the **algebraic** techniques from Chap 01, calculate the derivative of $f(x) = \frac{1}{x}$ at $x=2$.

$$f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}.$$

$$f'(2) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1}{2+h} - \frac{1}{2} \right) = \lim_{h \rightarrow 0} \left(\frac{2 - (2+h)}{2h(2+h)} \right) = \lim_{h \rightarrow 0} \frac{-h}{2h(2+h)}.$$

cancel h . We get

$$f'(2) = \lim_{h \rightarrow 0} \frac{-1}{2(2+h)} = -\frac{1}{4}.$$